

Total Appointments

5000

Total No-Shows

2423

No-Show Rate

51.15%

Attended Appointments

2314

Cancelled Appointments

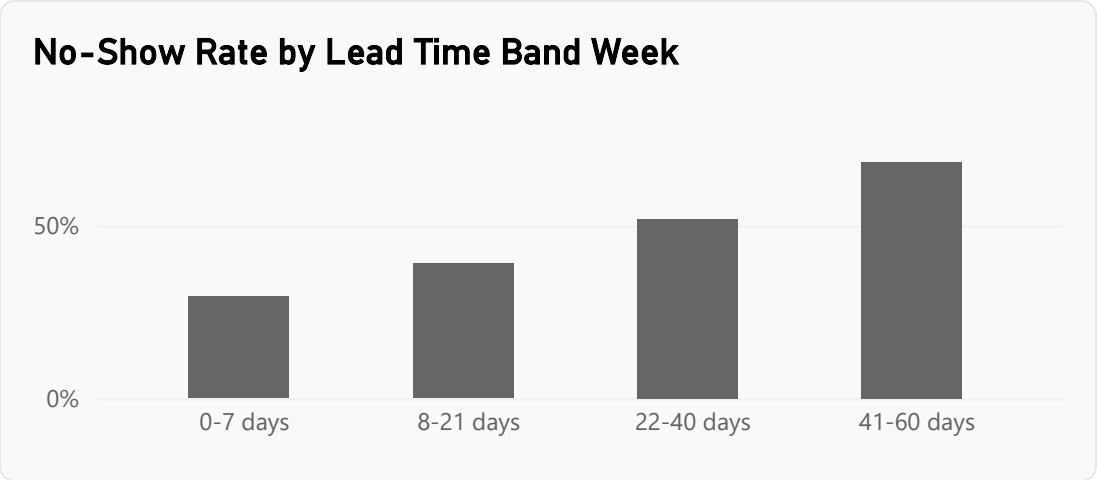
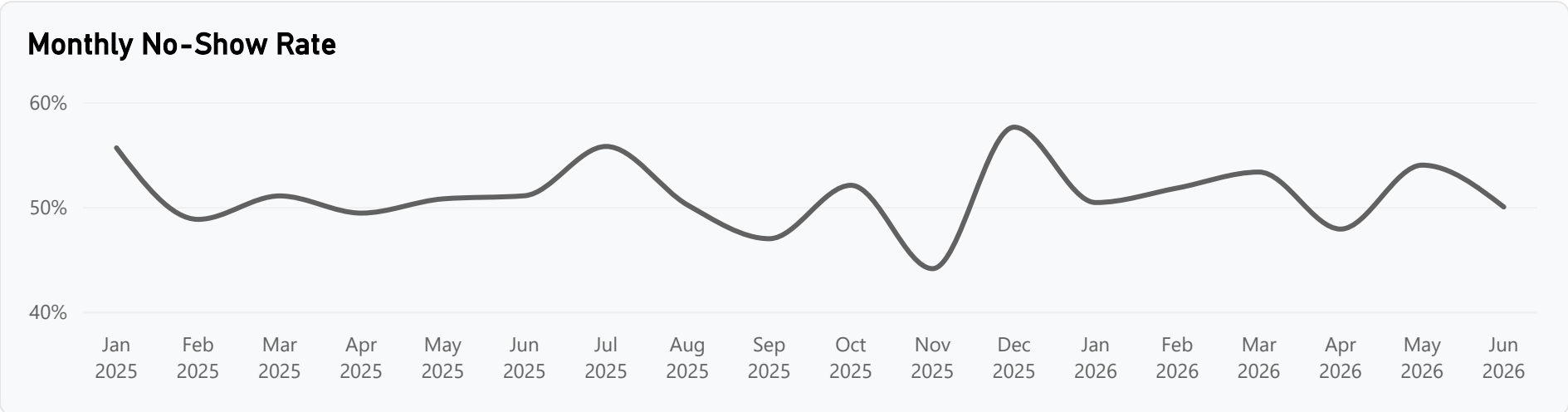
263

Attendance Rate

48.85%

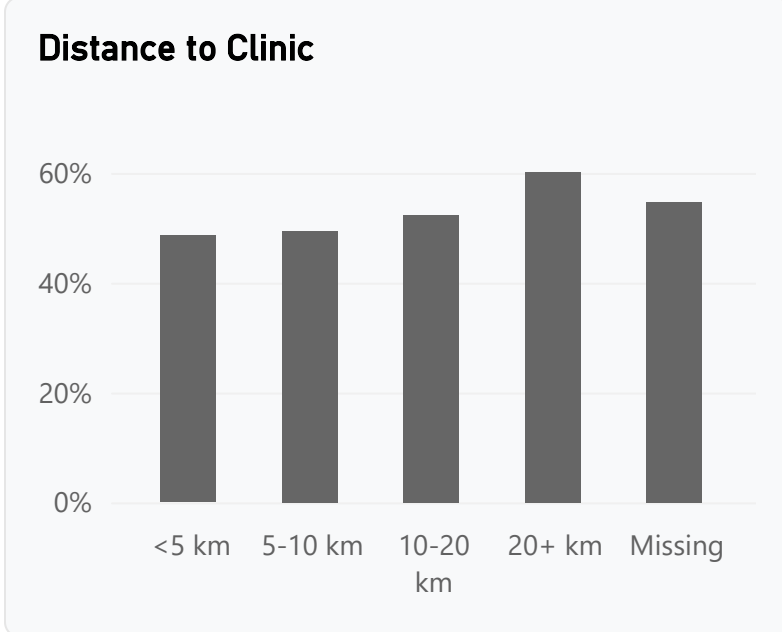
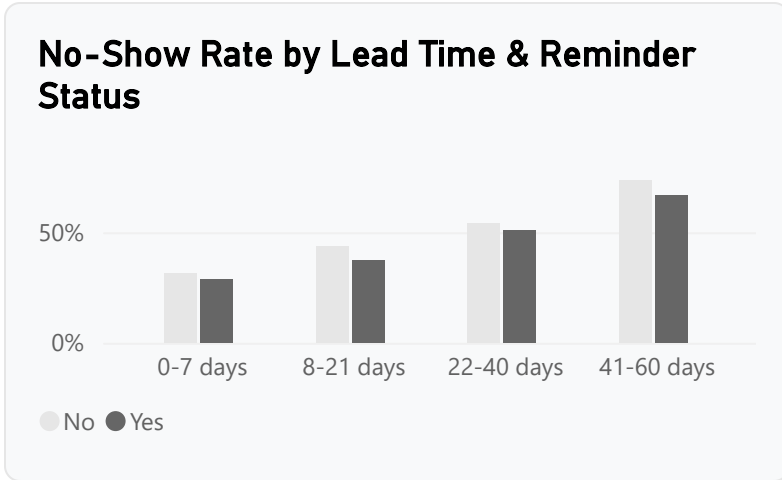
Eligible Appointments

4737



No-Show Risk: Lead Time * Previous History

Previous No-Show History	0-7 days	8-21 days	22-40 days	41-60 day
No Previous No-Show	22.82%	35.89%	48.19%	62.83%
Previous No-Show	38.96%	43.76%	56.78%	75.72%
Total	29.47%	39.02%	51.82%	68.49%



app_day

☐ Monday

☐ Tuesday

app_time

☐ Afternoon

☐ Evening

☐ 0-7 days

☐ 8-21 days

☐ 22-40 days

☐ 41-60 days

Previous No-Show Segment

☐ <2 previous no-shows

☐ 2+ previous no-shows

Key Insights

Distance matters. Patients living 20+ km from the clinic miss their appointments 60.2% of the time. This geographic risk factor is a much stronger predictor of no-shows than patient age or the day of the week.

The no-show rate jumps to a critical 83.3% when long booking lead times (41–60 days) combine with a history of 2+ previous no-shows.

No-show rates rise substantially as booking lead time increases, reaching approximately 68.5% for appointments booked 41–60 days ahead.

Reminder-associated appointments recorded lower observed no-show rates within each lead-time group.

Priority No-Show Segments

Segment	No-Show Rate	Priority
41-60 days + 2+ previous no-shows	83.3%	Critical
41-60 day lead time	68.5%	High
2+ previous no-shows	63.5%	High
20+ km from clinic	60.2%	High
Any previous no-show	57.8%	High
No reminder	54.6%	Moderate

Advanced Analysis & Data Science integration

Model ROC-AUC

68.9%

Model Recall

65.8%

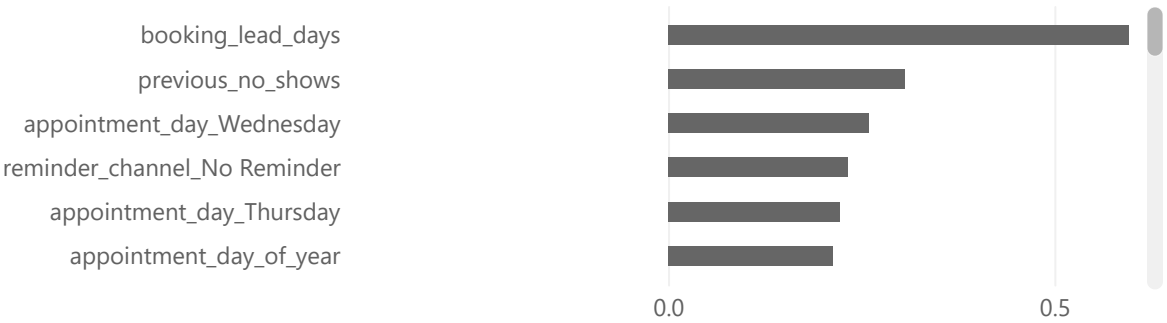
Model Precision

63.0%

Model F1

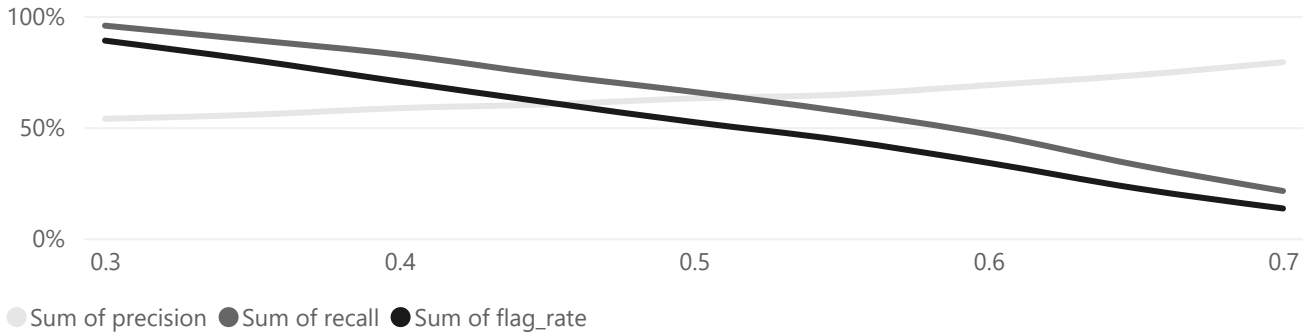
64.4%

Logistic Regression Coefficients



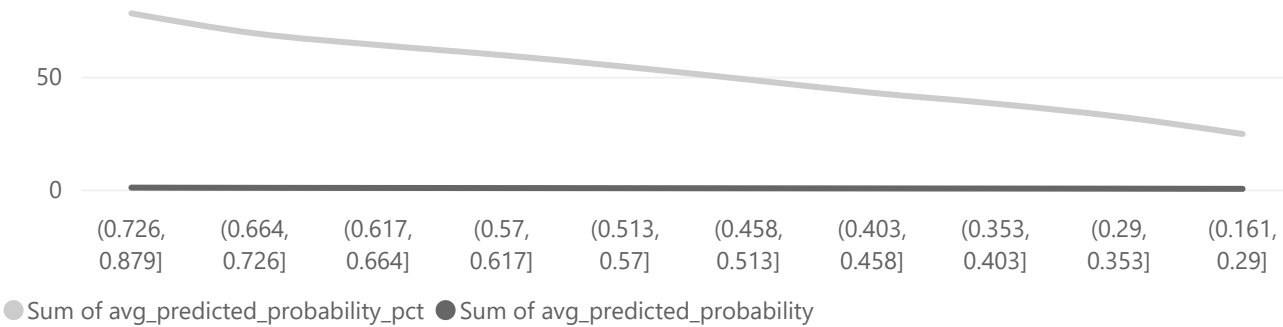
Positive coefficients are associated with higher predicted no-show probability; negative coefficients are associated with lower predicted probability, subject to model encoding/reference categories.

Classification Threshold Trade-Off



Lower thresholds capture more potential no-shows but require more outreach. Higher thresholds target fewer appointments but miss more true no-shows. Final threshold selection requires clinic capacity and cost assumptions.

Predicted Probability vs Actual No-Show Rate



Actual no-show rates rise from approximately 16.5% in the lowest score decile to 77.3% in the highest, indicating useful risk ranking.

Analytics * Data Science Evidence

Analytics Evidence	Model Evidence	Interpretation
Lead time: 29.5% → 68.5%	Strong predictive contribution	Strongest shared risk indicator
Previous history: 57.8% vs 46.3%	Positive predictive contribution	Prioritise supportive outreach
20+ km: about 60.2%	Distance contributes	Access burden deserves attention
No reminder: 54.6% vs 49.9%	Reminder variables contribute	Test reminder strategy prospectively
Follow-up: about 54.2%	Modest predictive contribution	Secondary / interacting factor

Integration limitation

The current scoring file contains model/test-row indices rather than the original HC-xxxxx appointment IDs. It has therefore not been joined to the main appointment dataset. A corrected scoring export containing the original appointment and patient identifiers has not been acquired for Week 7 row-level error analysis.